

AFRIN DANGE

✉ dangeafrin@iitb.ac.in | 📞 (+91) 93266 54927 | 📄 afrin-dange | 🌐 AfrinDange | 📦 afrindange.github.io

EDUCATION

Indian Institute of Technology Bombay

MS by Research | Artificial Intelligence & Data Science

Thesis: In-context Learning and Adaptation of Foundation Models for Time Series 📄

Advisor: Prof. Sunita Sarawagi

Jul '22 – Jun '25

CPI: 8.97/10

University of Mumbai

BE | Computer Engineering

Jul'18 – May '22

CPI: 9.61/10

RESEARCH EXPERIENCE

Amazon Lab126 | Research Scientist

Oct '25 – Present

Advisor(s): Dr. Deepak Gupta, Sankalp Dayal

- Identified limitations of evolution mechanisms in long-term memory through a systematic study of existing architectures, and validated the hypothesis by designing controlled experiments.
- Developed a taxonomy of episodic memory failures by annotating memory traces across SOTA memory systems and varying LLM scales using Thematic Analysis.
- Designed a graph-based provenance framework for tracing the origin and propagation of facts within memory systems, enabling detection of the stage at which memory failures occur.
- Designed process-level rewards for memory retrieval policy using provenance graph, improving QA accuracy by 7% when training a Qwen3-4B-Instruct based policy over outcome reward baselines such as Memora.

Indian Institute of Technology Bombay | Research Assistant

May '23 – Jul '25

Advisor: Prof. Sunita Sarawagi

Covariate-Aware Adaptation for Forecasting Foundation Models

- Designed a two-stage instance-level adaptation to equip foundation models for forecasting with covariates.
- Generated pseudo-forecasts to train a Gaussian Process adapter, enabling plug-and-play integration with foundation models.
- Benchmarked the adaptation method against state-of-the-art foundation models and supervised baselines, achieving a 24–27% improvement in forecast accuracy.

In-context Learning for Regression

Co-advisor: Dr. Praneeth Netrapalli (Google DeepMind)

- Analyzed loss convergence of transformer variants on regression tasks to identify mechanisms facilitating in-context learning.
- Proposed a modified causal attention mechanism enabling in-context learning with a single transformer layer.
- Improved length generalization through position-weighted loss and sequence-length distillation.

ICL-enabled Foundation Model for Forecasting

Co-advisor: Dr. Praneeth Netrapalli (Google DeepMind)

- Designed a hybrid transformer architecture for zero-shot time-series forecasting using in-context learning.
- Achieved zero-shot forecasting performance comparable to supervised baselines.
- Built a distributed pretraining pipeline using PyTorch Fully Sharded Data Parallel (FSDP).

PUBLICATIONS

- [1] **Afrin Dange**, Deepanshu Pandey, Rahul Rahaman, Sankalp Dayal, Deepak Gupta, *MnemeScope: Diagnosing Episodic Memory Failures and Designing Process Rewards for Learning Memory Policies*, Manuscript in preparation
- [2] **Afrin Dange**, Rahul Rahaman, Sankalp Dayal, Deepak Gupta, *Position: Task-Triggered Memory Evolution Can Destroy Memories with High Long-Term Utility*, COLM Workshop on Context Beyond Window, 2026

- [3] **Afrin Dange**, Sunita Sarawagi, *TFMAdapter: Lightweight Instance-Level Adaptation of Foundation Models for Forecasting with Covariates*, CIKM, ACM International Conference on Information and Knowledge Management, 2025 (Oral Presentation) [🔗](#)
- [4] **Afrin Dange**, Vaibhav Raj, Praneeth Netrapalli, Sunita Sarawagi, *Transformer Designs for In-context Learning in Foundation Models for Time Series Forecasting with Covariates*, ICML Workshop on Theoretical Foundations of Foundation Models, 2024 [🔗](#)

PROFESSIONAL SERVICE

- **Reviewer (Conference)**: NeurIPS 2026, ICLR 2026, ICLR 2025 (topics: in-context learning, foundation models, time series)
- **Reviewer (Workshop)**: ICML 2026, 2025 Workshop on Foundation Models for Structured Data, ICLR 2026 Workshop on Time Series in the Age of Language Models

TEST SCORES

TOEFL iBT 5.5/6.0 | Reading: 5.5, Listening: 5.5, Speaking: 4.5, Writing: 5.5 2026

SELECTED ACADEMIC PROGRAMS

Jul 2025 Armenian LLM Summer School, Yerevan, Armenia [🔗](#)
 Jan 2025 Google DeepMind Research Symposium, Bangalore, India [🔗](#)
 Jan 2025 Reinforcement Learning Workshop, IISc Bangalore, India [🔗](#)

TEACHING AND MENTORING

Graduate Teaching Assistant | IIT Bombay Spring '25
 CS726: Advanced Machine Learning

- Designed a lab assignment on LLM inference-time algorithms and developed an autograder, invigilated weekly quizzes, and evaluated homework and final exams.

Department Coordinator | IIT Bombay Jul '23 – Jun '24
 Institute Student Companion Programme (ISCP)

- Organized department-level induction under ISCP; led a team of three student mentors and personally mentored two incoming graduate students throughout their first year.

INVITED TALKS

Conditional Generation Using GANs and Diffusion Models Aug '24
 ISTE Program at VESIT | Mumbai, India

Generative AI for Vision Feb '24
 ISTE Program at FCRIT | Mumbai, India

RELEVANT COURSEWORK

Mathematics

Introduction to Probability & Random Processes; Applied Linear Algebra; Optimization in Machine Learning

Machine Learning

Foundations of Machine Learning; Advanced Machine Learning; Speech, NLP and the Web; Deep Learning for NLP; Information Retrieval & Mining for Hypertext & the Web; Organization of Web Information; Human-Centered AI; Introduction to AI, Data & Policy; Advanced LLM Agents (UC Berkeley [🔗](#))